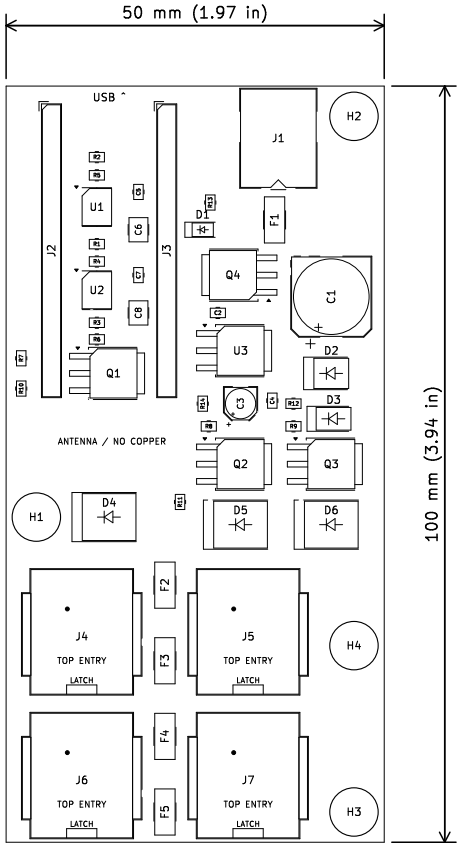


COMPONENT LEGEND

Reference	Component / function	Value / key specification
J1	DC barrel power input	12 V, centre +; 8 A; 5.5 x 2.1 mm
J2, J3	Nano female sockets	15-way; 2.54 mm pitch
J4-J7	Vertical RJ11 RGB outputs	Top-entry 6P4C; 1.5 A/contact
F1	Input fuse	6.3 A / 125 V fast
F2-F5	RJ11 branch fuses	3.15 A / 125 V fast
U1, U2	Dual MOSFET gate drivers	UCC27524ADR
U3	Linear voltage regulator	L78M09CDT-TR; 9 V
Q1-Q3	RGB low-side N-channel MOSFETs	AOD4184A
Q4	Reverse-polarity P-channel MOSFET	AOD4185
D1	Gate-protection Zener diode	12 V; BZT52C12-7-F
D2	Input surge suppressor (TVS)	SMBJ15A
D3	Nano VIN isolation Schottky diode	SS14-E3/61T
D4-D6	RGB cable-clamp Schottky diodes	SSC54-E3/57T
C1	Bulk input capacitor	470 µF / 35 V
C2	Regulator input bypass capacitor	330 nF / 50 V
C3	Regulator output capacitor	10 µF / 25 V
C4, C5, C7	Regulator / driver bypass capacitors	100 nF / 50 V
C6, C8	Driver local supply capacitors	4.7 µF / 50 V
R1-R3	PWM input series resistors	1 kΩ
R4-R6, R10-R12	Input / MOSFET-gate pulldown resistors	100 kΩ
R7-R9	MOSFET-gate series resistors	33 Ω
R13	Reverse-polarity gate bias resistor	10 kΩ
R14	Regulator minimum-load resistor	1.8 kΩ
H1-H4	M3 mounting holes (not fitted parts)	non-plated 3.2 mm hole

Exact manufacturer part numbers: pcb-bom.csv.

All carrier parts fitted. Nano supplied separately. J4-J7: vertical RJ11, 1.5 A/contact.
50 x 100 mm / FR4 1.6 mm / 2 oz both sides / ENIG

Sheet:
File: pcb.kicad_pcb

Title: 12 V RGB carrier – top assembly / plug-in Nano 33 IoT

Size: A4	Date: 2026-10-12	**Rev: C**
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